

The Linear Model (LM)

Key ideas in 15 minutes

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The linear model, in one picture

- ▶ y is a draw from a Normal distribution
- ▶ its mean is a **weighted sum** of the predictors
- ▶ its standard deviation σ is **constant**

$$y_i \sim \mathcal{N}(\underbrace{\alpha + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i}}_{\mu_i}, \sigma)$$

Why should you care?

- ▶ LMs (& friends: t -tests, ANOVA, GLMs, GLMMs) are the workhorse of our field(s)
- ▶ Fitting one means estimating $\hat{\alpha}, \hat{\beta}_1, \dots, \hat{\beta}_k, \hat{\sigma}$
- ▶ Everything else we'll cover this semester builds on this model

Four core assumptions

- ▶ **Independence** of observations
- ▶ **Normality** of the residuals
- ▶ **Equal variance** of the residuals (homoscedasticity)
- ▶ **Additivity & linearity** of effects

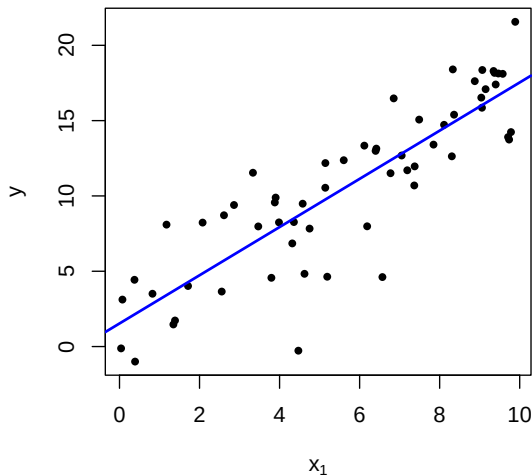
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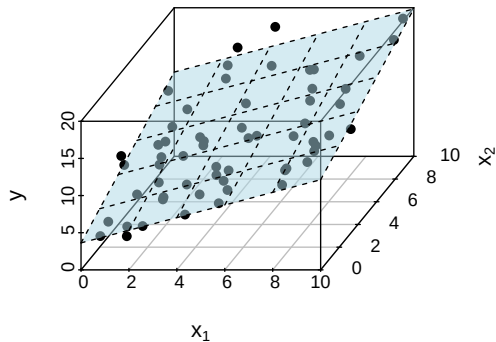
Violations of *independence*, *normality*, or *homoscedasticity* can invalidate your inferences.
Violations of *linearity* are often (partially) fixable by transforming predictors.

Geometrically: a line, a plane, a hyperplane

One predictor



Two predictors



With k predictors: a hyperplane in $k + 1$ dimensions.

What makes a line fit well?

Residual sum of squares

$$RSS = \sum_i (y_i - \hat{y}_i)^2$$

smaller RSS $\rightarrow$ better fit

Likelihood

$$\Lambda = \sum_i \log p(y_i \mid \hat{\alpha}, \hat{\beta})$$

higher $\Lambda \rightarrow$ better fit

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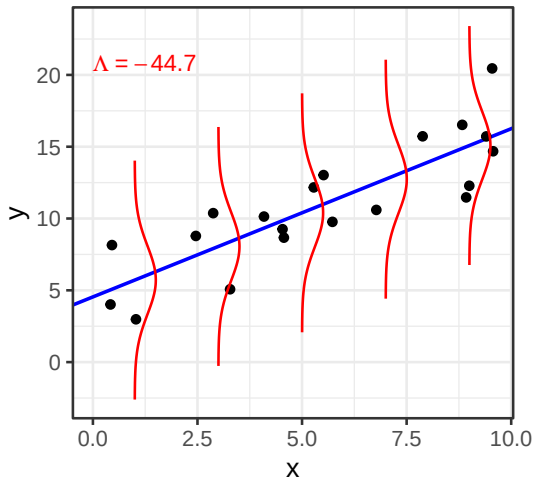
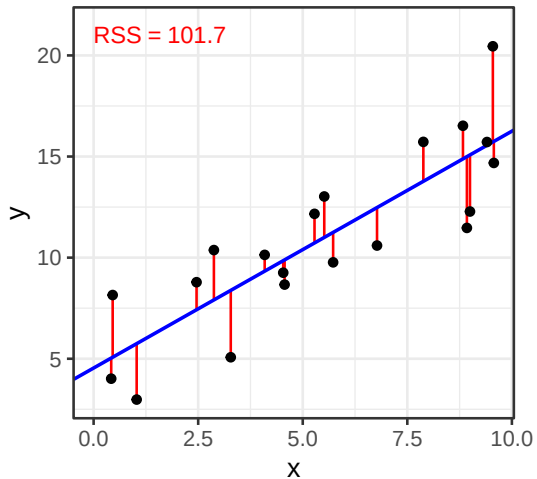
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For the LM, minimizing $RSS \equiv$ maximizing Λ .

Same idea, side by side



A worked example

- ▶ Data: reaction times (RT) in a 2AFC task, as a function of VOT
- ▶ One participant, 72 trials
- ▶ Question: does VOT *predict* RT?

Fitting the LM

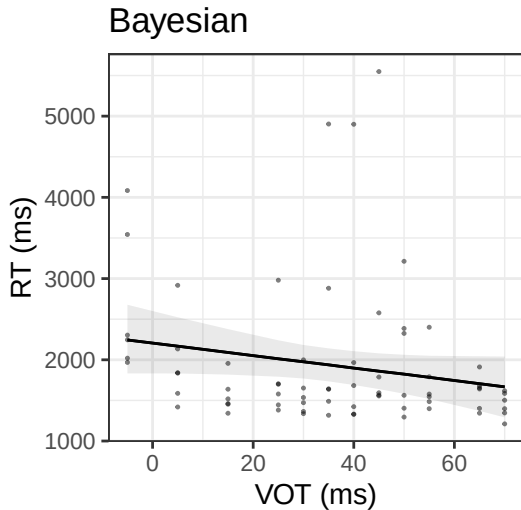
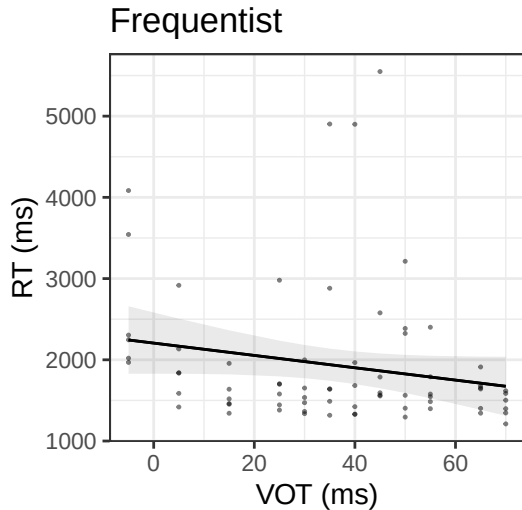
```
lm.frequentist <- lm(Response.RT ~ 1 + VOT, data = dd)
coef(summary(lm.frequentist))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2205.812471	191.952542	11.491447	8.998604e-18
VOT	-7.594767	4.556115	-1.666939	9.999508e-02

► $\hat{\beta}_{VOT} = -7.6$, $t = -1.67$, $p = 0.1$

► Not significant here

Frequentist $\approx$ Bayesian (for this example)



Reading the output

- ▶ **Coefficients:** $\hat{\beta}$, its uncertainty, and (frequentist) a t/p -value
- ▶ **Residual SD:** $\hat{\sigma}_\epsilon$ (how noisy is y around the line?)
- ▶ R^2 : how much variance is explained
- ▶ Bayesian: no p -value—instead, the full posterior (e.g., % of samples < 0)

Writing it up

“We used linear regression to regress RT against VOT. The effect of VOT was not significant ($\hat{\beta} = -7.6$, $t = -1.67$, $p \geq .1$).”

- ▶ Report the estimate **and** a plain-language interpretation
- ▶ Use $\hat{\beta}$ (not β) — it's an *estimate*
- ▶ Don't equate “significant” with “large effect”

Writing it up (Bayesian)

“We used a Bayesian linear regression to regress RT against VOT, with weakly regularizing priors ($\mathcal{N}(0, 2.5)$). There was moderate evidence that RTs decreased with increasing VOT ($\hat{\beta} = -7.6$, 95%-CI = $[-16.7, 1.7]$; $BF = 17.9$).”

- ▶ Report the **posterior** (e.g., its mean/median and credible interval)
- ▶ No p -value — instead, summarize the posterior directly (e.g., % < 0 , an evidence ratio/Bayes factor)
- ▶ Avoid dichotomous (“significant”/“not significant”) language

Take-aways

- ▶ LM: $y \sim \mathcal{N}(X\beta, \sigma)$ — a line/plane/hyperplane + normal noise
- ▶ Fitting = minimizing RSS = maximizing likelihood Λ
- ▶ Same model, two frameworks (frequentist / Bayesian) — often similar answers
- ▶ Next: predictor coding & interactions. Then LLMs & GLMMs.